2 (c) Posterror PDF of R if the offshore has been damaged we have,

$$f''(\theta_i) = \frac{P(\epsilon|\theta) f'(\theta)}{\int_{-\infty}^{\infty} P(\epsilon|\theta) f'(\theta) d\theta}$$

$$= K \, L(\theta) f'(\theta)$$

Here,

$$L(r) = \frac{r^2}{900.00} \quad , f_R'(r) = \left(\frac{a}{r} - \frac{a}{40}\right) \qquad 10 \le r \le 40$$

$$L(\theta) = \left[\int_{-\infty}^{\infty} \frac{r^2}{90000} \left(\frac{a}{r} - \frac{a}{40}\right) dr\right]^{-1}$$

$$= \left[\int_{10}^{40} \frac{r^2}{90000} \left(\frac{a}{r} - \frac{a}{40}\right) dr\right]^{-1}$$

$$= \left[\frac{a}{400}\right]^{-1}$$

$$\therefore f_R''(r) = \frac{\frac{r^2}{90000}\left(\frac{a}{r} - \frac{a}{40}\right)}{\frac{a}{400}} = \frac{r^2}{225}\left(\frac{1}{r} - \frac{1}{40}\right)$$

$$\therefore f_R''(r) = \begin{cases} \frac{r^2}{225}\left(\frac{1}{r} - \frac{1}{40}\right) & 10 \le r \le 40 \\ 0 & r > 40 \end{cases}$$

3 (b) Mean and Variance of X_1 and X_2,

$$E[X] = \sum_{\text{all } x_i} x_i \, p_X(x_i)$$

$$\Rightarrow E(X_1) = 0 \times \frac{5}{21} + 1 \times \frac{7}{21} + 2 \times \frac{9}{21} = 1.1905 = 25/21$$

Also, $$E[g(x)] = \sum_{\text{all } x_i} g(x_i)\, p_X(x_i)$$

$$\Rightarrow E[X_1^2] = \sum_{\text{all } x_{1i}} x_{1i}^2 \, p_{X_1}(x_{1i})$$

$$= 0^2 \times \frac{5}{21} + 1^2 \times \frac{7}{21} + 2^2 \times \frac{9}{21}$$

$$= 43/21$$

$$\Rightarrow \text{Var}[X_1] = E[X_1^2] - (E[X_1])^2$$

$$= 43/21 - (25/21)^2 = \frac{278}{441} = 0.6304$$

Similarly,

$$E[X_2] = 2 \times 9/21 + 3 \times 12/21 = 18/7 = 2.5714$$

$$E[X_2^2] = 2^2 \times \frac{9}{21} + 3^2 \times \frac{12}{21} = \frac{48}{7} = 6.8571$$

$$\cancel{\text{Var}[X_2] = \frac{48}{7} - \left(\frac{18}{7}\right)^2 = \frac{12}{49} = 0.2449}$$

$$\text{Var}[X_2] = E[X_2^2] - (E[X_2])^2 = \frac{48}{7} - \left(\frac{18}{7}\right)^2$$

$$= \frac{48}{7} - \frac{324}{49} = \frac{12}{49} = 0.2449$$

4

(b) The second order mean

we have,

for $Y = g(X)$

second order approximation is,

$$E(Y) \approx g(\mu_X) + \frac{1}{2} Var(X) \left.\frac{d^2 g}{dX^2}\right|_{\mu_X}$$

Here,

$$\cancel{E(D) \approx \frac{5 \times \left(\frac{1000}{12}\right) \times 180^4}{384}}$$

$$E(D) \approx \frac{5 w l^4}{384 \mu_E I} + \frac{1}{2} Var(E) \left.\frac{d^2 D}{dE^2}\right|_{\mu_E}$$

$$\frac{d^2 D}{dE^2} = \frac{d^2}{dE^2}\left(\frac{5wl^4}{384EI}\right) = \frac{5wl^4}{384I} \frac{d^2}{dE^2}\left.\left(\frac{1}{E}\right)\right|_{\mu_E}$$

$$= \frac{5wl^4}{384I} \left.\frac{2}{E^3}\right|_{\mu_E} = \frac{5wl^4}{384I}\left(\frac{2}{\mu_E^3}\right)$$

$$\therefore E(D) \approx \frac{5wl^4}{384\mu_E I} + \frac{1}{2} Var(E) \frac{5wl^4}{384I} \frac{2}{\mu_E^3}$$

$$= 0.4394 + \frac{1}{2}(750000)^2 \frac{5 \times \frac{1000}{12} \times 180^4 \times (2)}{384 \times 864 \times (3000000)^3}$$

$$= 0.46692 \text{ m}$$

4

(c) If E is lognormal with parameters λ_E, ξ_E^2 then $\ln E$ is normally distributed as $N(\lambda_E, \xi_E^2)$

Here,

$$D = \frac{5wl^4}{384EI} = \frac{C}{E} \quad \text{where, } C = \frac{5wl^4}{384I}$$

$$C = \frac{5 \times \frac{1000}{12} \times 180^4}{384 \times 864}$$

$$= 1318359.375$$

$$D = C/E$$

taking log on both sides,

$$\ln D = \ln C/E = \ln C - \ln E = (-1)\ln E + \ln C$$
$$\equiv Y = aX + b$$

Since $\ln E$ is normally distributed, $\ln D$ is also normally distributed. Thus D is lognormal.

Lognormal Parameters for E:

$$\xi_E^2 = \ln(1 + \delta_E^2)$$

$$= \ln(1 + 0.25^2) = 0.060625$$

$$\lambda_E = \ln \mu_E - \frac{1}{2}\xi_E^2 = \ln(3000\,000) - \frac{1}{2}\times 0.060625$$

$$= 14.88381D$$

$\therefore$ $E \Rightarrow \text{lognormal}(\lambda_E, \xi^2) = \text{lognormal}(14.88, 0.0606)$

$\ln E \Rightarrow N(\lambda_E, \xi^2) : N(14.88, 0.06062)$

lognormal parameters for D,

for a linear function $Y = aX + b$,

$$E(Y) = a\,E(X) + b$$
$$Var(Y) = a^2\,Var(X)$$

for $\ln D = (-1)\ln E + \ln C$

$$\xi_D^2 = Var(\ln D) = (-1)^2\,Var(\ln E) = \xi_E^2 = 0.060625$$

$$\lambda_D = \cancel{E(D)}\ E(\ln D) = (-1)\,E(\ln E) + \ln C$$
$$= (-1)\,14.88381 + \ln(1318359.375)$$
$$= -0.791912$$

$\therefore$ $\ln D \Rightarrow N(-0.7919, 0.0606)$

$D \Rightarrow \text{lognormal}(-0.7919, 0.0606)$

we have, for lognormal Distribution,

$$f_X = \frac{1}{\sqrt{2\pi}\,(\xi x)}\ \exp\left[-\frac{1}{2}\left(\frac{\ln x - \lambda}{\xi}\right)^2\right] \qquad x \geq 0$$

$$\therefore f_D(d) = \frac{1}{\sqrt{2\pi}\,\xi_D d} \exp\left[-\frac{1}{2}\left(\frac{\ln d - \lambda_D}{\xi_D}\right)^2\right], \quad d \geq 0$$

$$\sqrt{2\pi}\,\xi_D = \sqrt{2\pi} \times \sqrt{0.060625} = 0.6172$$

$$\Rightarrow f_D(d) = \frac{1}{0.6172d} \exp\left[-\frac{1}{2}\left(\frac{\ln d + 0.7919}{0.24622}\right)^2\right], \quad d \geq 0$$

$$= \frac{1}{0.6172d} \exp\left[-\frac{(\ln d + 0.7919)^2}{0.12125}\right], \quad d \geq 0$$

Exact Mean And coefficient of variation:

we have,

$$\xi_D^2 = \ln(1 + \delta_D^2) \Rightarrow \delta_D = \left(e^{\xi_D^2} - 1\right)^{1/2}$$

$$\therefore \delta_D = \left(e^{0.060625} - 1\right)^{1/2} = 25\%$$

$\therefore$ exact coefficient of variation of D = 25%.

Also,

$$\lambda_D = \ln \mu_D - \frac{1}{2}\xi_D^2 \Rightarrow \mu_D = \exp\left(\lambda_D + \frac{1}{2}\xi_D^2\right)$$

$$\Rightarrow \mu_D = \exp\left(-0.791912 + \frac{1}{2} \times 0.060625\right)$$

$$= 0.46692 \text{ m}$$

$\therefore$ exact mean = 0.46692 m